Introduction to Robotics, Lecture 4

Differential Kinematics
Joint and Tool Velocities

MIPT

Center for Cognitive Modeling

Differential Kinematics

Differential kinematics relates joint velocities to the linear and angular velocities of a robot's end effector. This relation is used to control motion, construct trajectories, and analyze singular configurations. The following three sources explain the topic in different ways. Start with the clearest explanation for you, then compare all three.

Course Material Robotics 1

Prof. Alessandro De Luca — Sapienza University of Rome.

- 11. Differential Kinematics — PDF

Introduction to Robotics — Stanford

Prof. Oussama Khatib — Stanford Engineering Everywhere.

- Lecture 6: The Jacobian and instantaneous kinematics
- Lecture 7: Jacobians and singularities

Modern Robotics — Chapter 5

Kevin M. Lynch and Frank C. Park — Northwestern University.

- Chapter 5.1.1: Space Jacobian
- Chapter 5.1.2: Body Jacobian

Reference Materials

- A. De Luca — 11 DifferentialKinematics
- Oussama Khatib — Differential kinematics
- Kevin M. Lynch and Frank C. Park — Space and body Jacobians

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